MATH SMASH:CARD ADVENTURE

Theme : Cards & Roguelike & Educational

Yuhao Liu - A01787782
Daniela Janet Gil Gonzalez - A01752908
Mariana Ramirez Cervera

ELEVATOR PITCH

Math Smash: Card Adventure is a roguelike deck-building game designed for children, where players solve math problems to activate cards and defeat enemies in a dynamic and strategic environment. Before each battle, players prepare their deck and strategy, while the outcome of combat depends on their ability to solve problems quickly and accurately under time pressure. Each run offers a unique experience due to variations in maps, enemies, and rewards.

BASIC GAME DESCRIPTION

In Math Smash: Card Adventure, players explore a procedurally generated map that changes in every run, including different paths, enemies, and objects, so that each playthrough feels unique and engaging.
Before each battle, players must carefully select and organize their deck, choosing from attack, defense, and skill cards, while preparing a strategy that will allow them to progress through increasingly difficult encounters and eventually defeat the final boss.

ROGUELiTE AND TRADiNG CARD GAME ELEMENTS

Roguelite and Trading Card Game Elements

- The map changes every run (not only enemies, but also paths and objects)
- Progressive difficulty across runs
- Meta-progression system: players retain certain elements (such as skill cards) after losing.

Trading Card Game Elements

- Deck selection before each battle
- Cards obtained as rewards from combat and objects
- Combinations between different card types
- Multiple strategies depending on deck composition

RULES AND CORE MECHANICS

The core mechanic of the game consists of activating cards by solving math problems within a limited time.

- Correct answer: full effect on the enemy
- Incorrect answer: reduced effect.

The game features three levels of difficulty to activate the cards:

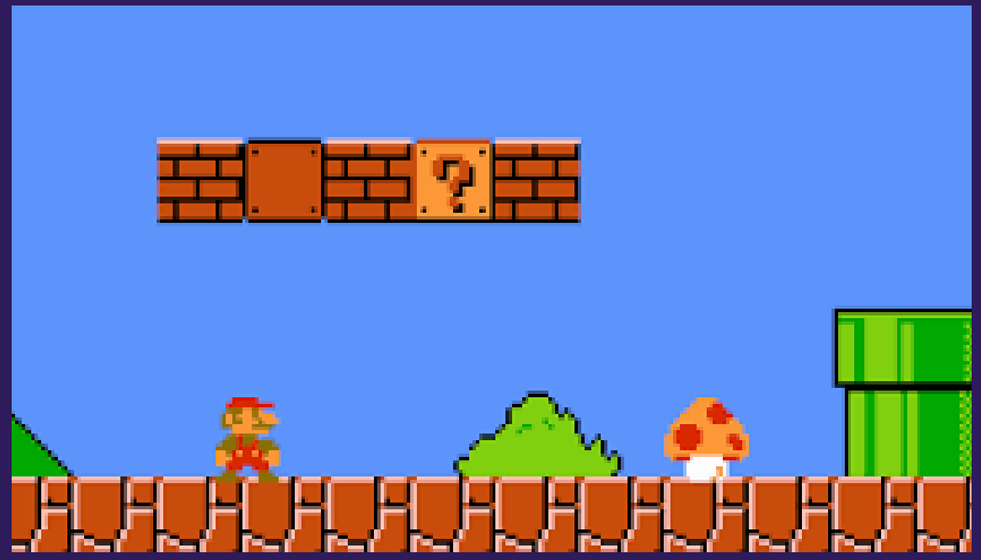
- Easy map: addition and subtraction
- Medium map: multiplication and division
- Hard map: mixed operations

Additionally:

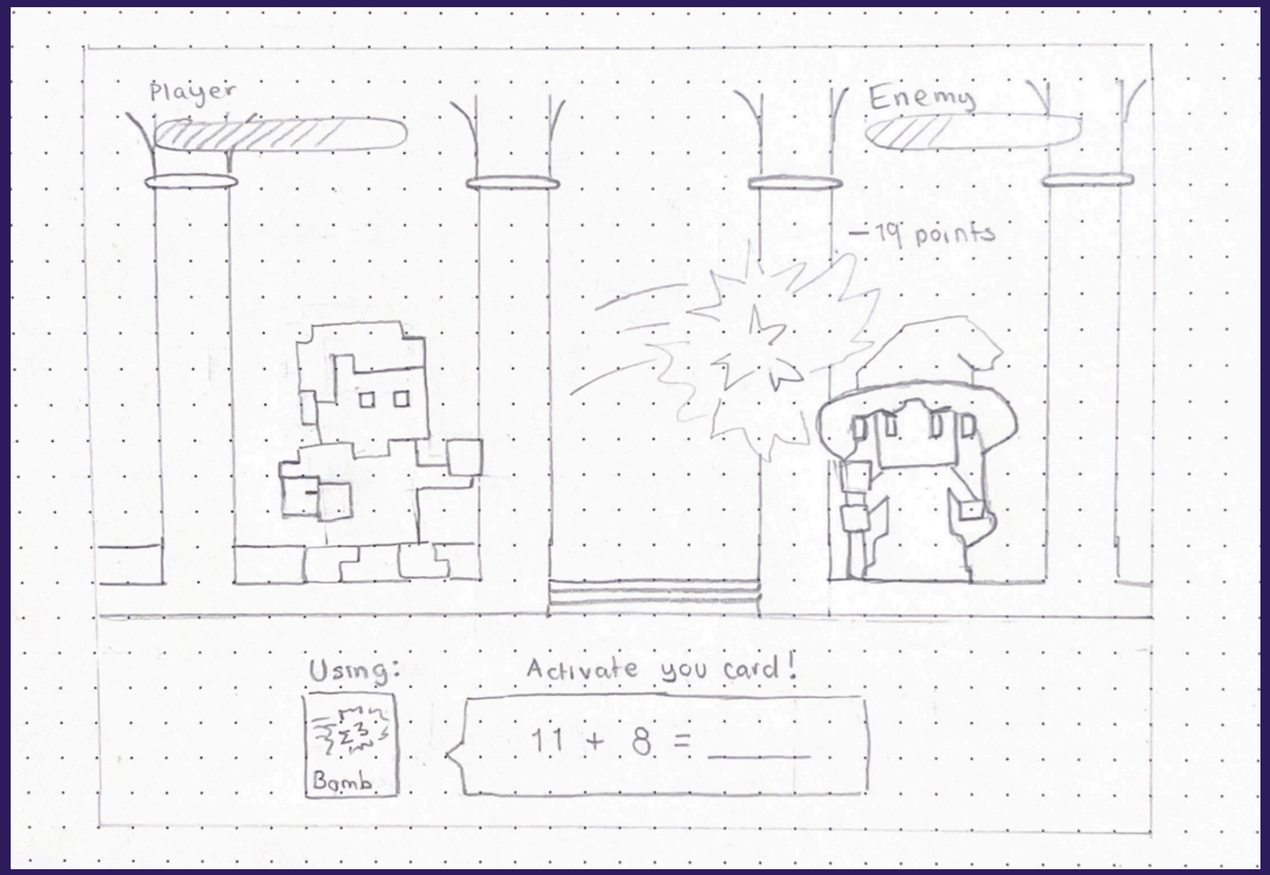
- The time available to answer decreases as difficulty increases

VISUAL STYLE OF THE GAME

The game features a colorful and accessible visual style designed to appeal to younger audiences while maintaining clarity and readability.
The visual design takes inspiration from Super Mario Bros., which uses bright colors and simple character design, as well as Card Wars: Adventure Time, known for its clear card layout and intuitive interface.
The user interface (UI) is designed to clearly display:
Available cards
Math problems
Player and enemy status
Visual feedback such as animations, color changes, and effects is used to improve engagement.



UI SKETCH



DATA AND STATISTICS COLLECTED BY THE GAME

Player Data

- Run duration
- Win/loss results
- Selected difficulty level (depending on the map)

Card Data

- Most frequently used cards
- Activation difficulty level depending on the map.
- Animations for each attack.

Combat Data

- Damage dealt and received
- Number of correct and incorrect answers

Progression Data

- Skills unlocked (the cards)
- Total combats completed
- Player improvement over time

THANKS FOR LISTENING!